MIL-S-13544 (Ord)
24 June 1954

MILITARY SPECIFICATION
SET, WIRING, M10A1E1

1. SCOPE

1.1 This specification covers a wiring set, which is a part of System, Local Control, M16A1E1, used with Gun, Twin 40MM, Self-propelled, M42.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and drawings, of the issue in effect on the date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Military	
MIL-I-12659	Inverter, 350 Volt Ampere, D7692245
MIL-S-13498	Solenoid, Firing, D7665749 and F8208586
U. S. Army	
51-7	Instruments, Fire Control; General Specification Governing the Manufacture and Inspection of

DRAWINGS

Ordnance Corps	
F7665897	Set, Wiring, M10A1E1

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.- Unless otherwise specified, the requirements common to all fire control instruments as contained in Specification 51-7 shall be mandatory.

3.2 Material.- Material shall be in accordance with the applicable drawings and specifications forming a part of this specification.

3.3 Performance

3.3.1 Inverter.- The inverter shall meet the requirements of Specification MIL-I-12659.

1/29

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3.3.2 Solenoid.- The solenoid shall meet the requirements of Specification MIL-S-13498.

3.3.3 Slip ring.- The Radio Frequency Slip Ring Assembly, Drawing D7665533, shall meet the following requirements:

3.3.3.1 Continuity.- All circuits shall maintain electrical continuity and shall have constant impedance during 360 degrees of rotation.

3.3.3.2 Data circuits.- Each data circuit lead shall be capable of carrying continuously a current of one ampere.

3.3.3.3 Power circuits.- The power circuit shall be capable of carrying continuously a current of 150 amperes, and in surges current up to 600 amperes.

3.3.4 Elevation switch.- The switch lever shall engage the slewing clutches of both oil gears prior to the closing of the elevation oil gears switch into the IN position.

3.3.5 Control selector switch.- The control selector switch box shall operate as specified.

3.4 Environmental

3.4.1 Temperature.- Components of the wiring set shall be operable throughout the air temperature range extending from plus 51.5°C (/ 125°F) (minimum exposure of 4 hours with full impact of solar radiation, 360 BTU/Ft. Sq/Hr.), to minus 31.6°C (- 25°F) (minimum exposure of 3 days without benefit of solar radiation). Components of the wiring set shall be capable of safe storage within the air temperature range extending from plus 71°C (/ 160°F) (for periods as long as 4 hours per day) to minus 62.2°C (- 80°F) (for periods of at least 3 days duration). Test shall be conducted in accordance with 4.2.2.

3.4.2 Humidity.- Components of the wiring set shall not be affected adversely by the humidity test specified in 4.2.3.

3.4.3 Vibration.- Components of the wiring set shall not be affected adversely by the vibration test specified in 4.2.4.

3.4.4 Shock.- Components of the wiring set shall not be affected adversely by the shock test specified in 4.2.5.

3.5 Workmanship.- Workmanship shall be in accordance with Specification 51-7.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.- Unless otherwise specified, each wiring set shall be inspected for conformance with this specification and Specification 51-7. Unless otherwise specified, standard ambient tests shall be performed at a temperature of 60°F to 90°F, at a barometric pressure of 28 to 32 inches of mercury, and a relative humidity not greater than 80 percent.

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4.2 Environmental

4.2.1 General.- The environmental tests specified shall be limited to those contained in the following paragraphs. Three of the first five of each component being tested plus one out of every additional 100 or fraction thereof shall be tested. Inverters and solenoids shall be excluded from these tests.

4.2.2 Temperature.- The component shall be exposed to an air temperature of minus 62.2°C (- 80°F) for a period of 3 days. The air temperature then shall be raised to minus 31.6°C (- 25°F) and after the internal temperature of the component has reached equilibrium (approximately 4 hours), it shall be operated in its intended manner. The air temperature shall be raised to plus 71°C (/ 160°F) and after the internal temperature of the component has reached equilibrium at that temperature it shall be allowed to remain at that temperature for a period of 4 hours. The air temperature shall be raised to plus 51.5°C (/ 125°F) and after the internal temperature of the component has reached equilibrium at that temperature the top surface shall be exposed to 360 BTU/Ft. Sq/Hr., for a period of 4 hours. The component then shall be operated in its specified manner.

4.2.3 Humidity.- A test chamber that is capable of being sealed shall be used; the source of heat shall be arranged in such a manner that the radiant heat will not fall upon the component being tested. A component that has been tested and found to be in conformance with the performance requirements of this specification shall be placed in the chamber, the temperature of which is adjusted to plus 46.1°C ± 1.1°C (/ 115°F ± 2°F) and the relative humidity adjusted to 95 ± 5 percent. The component shall remain under the above condition for a period of 6 hours. The temperature of the chamber shall be readjusted to plus 26.1°C ± 1.1°C (/ 80°F ± 2°F) and the component shall be allowed to remain under this new condition for a period of 6 hours. The transitional period shall be 2 hours; the rate of change during that period shall be uniform. The above cycle shall be repeated 10 times. Immediately after this test the component shall meet the performance requirements of this specification. Only slip ring assemblies, elevation switches, and control selector switch boxes shall be subjected to the humidity test.

4.2.4 Vibration.- Components shall be mounted, using their own mounting provisions, to simulate field use. Each unit tested shall be vibrated successively along each of the 3 mutually perpendicular directions with accelerations and excursions as specified below:

Frequency limits	Excursion	Acceleration	Time (each direction)
0 - 10 cps	0.060 inch	- - - - -	1 hour
10 - 20 cps	0.030 inch	- - - - -	2 hours
20 - 400 cps	- - - - -	4.5 g's	2 hours

4.2.5 Shock.- Components shall be mounted, using their own mounting provisions, to simulate field use. Each unit tested shall be subjected to 20 g's of shock in each direction along each of the 3 mutually perpendicular directions, making a total of 18 shocks. Each shock impulse shall have a time duration of 11 ± 1

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milliseconds; the maximum g value shall occur at approximately 5.5 milliseconds and shall be within ± 10 percent. Filters used shall have cut-off frequencies selected to eliminate all harmonics above the fundamental, from the measured shock impulse.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with instructions forming a part of the contract or furnished by the contracting officer in connection with the contract.

6. NOTES

6.1 Ordering data.- Contracts and invitation for bids should specify the title, number, date of this specification and type of packing.

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